Assignment 3 - Discrete Mathematics (Monsoon 2025)

"A great discovery solves a great problem but there is a grain of discovery in any problem."
- by George Pólya

Submission deadline: 11:59 PM on 23rd November, 2025. **Maximum marks**: 40

Instructions. Consider the following while preparing the answers for the given questions.

1. Kindly go over the academic dishonesty policy of IIIT Delhi given on this link. Any deviation from this policy will lead to the strictest possible punishment.
2. Each question is of 10 marks.
3. It is prohibited to use AI-related tools like chatGPT to solve the questions, unless explicitly mentioned. Any violation in this regard will be handled very strictly.
4. You are encouraged to write solutions in LaTeX. Anyone who submits homework in LaTeX(i.e., in the pdf format and not the source code) will get <u>an additional 20% of their score in that homework</u>.
5. While writing the solutions, you can refer to the topics, theorems, examples, etc., taught in the class. It is not allowed to refer to anything related to propositional and predictive logic not covered in the class.
6. Late submissions are not allowed.
7. If you discuss anything related to the solution of a question in this assignment with anyone, you must mention their names.

1 Problem Set

Question 1.1 (10 marks). *Let A be the set of all finite sequences in $\mathbb{N}$. Is A countably infinite or uncountable? Prove your answer.*

A finite sequence in $\mathbb{N}$ is a sequence having finitely many natural numbers. For example, $(1,2,3), (9,100,0,5), (1)$.

Question 1.2.

1. *(5 marks). Let $n \in \mathbb{N} - \{0\}$ and A be the set of all $0,1$ sequences containing n bits. Let R be a relation on A defined as follows: for every $(a_1, a_2, \ldots, a_n), (b_1, b_2, \ldots, b_n) \in A$, $((a_1, \ldots, a_n), (b_1, \ldots, b_n)) \in R$ if and only if*
$$a_1 + \cdots + a_n = b_1 + \cdots + b_n.$$
Prove that R is an equivalence relation. Count the number of equivalence classes in A/R. Describe an equivalence class in A/R that has the maximum number of elements among all equivalence classes in A/R, and count the number of elements present in this equivalence class.

2. *(5 marks). Let $A = \{1, \ldots, n\}$ for $n \in \mathbb{N} - \{0\}$. Let B be all permutations on A. Define the equivalence relation R on B as follows: for every $f, g \in B, (f, g) \in R$ if and only if there exist $\pi, \sigma \in B$ such that $f = \sigma \circ g \circ \pi$.*
   (a) *Let $[f]_R$ be an arbitrary equivalence class in B/R. Count the number of elements in $[f]_R$.*
   (b) *Count the number of equivalence classes in B/R. Justify your answers.*

Question 1.3.

1. *(5 marks). Let $\{A_i : i \in \mathbb{N}\}$ be the set of countably infinite sets, i.e., for every $i \in \mathbb{N}, A_i$ is countably infinite. Prove that for every $n \in \mathbb{N}, n \geq 1$, $A_1 \cup \cdots \cup A_n$ and $A_1 \cap \cdots \cap A_n$ are countable.*

2. *(5 marks). Prove the following for every $n \in \mathbb{N}, n \geq 1$: Let A be a set of n elements and B be the set of all subsets of A containing even number of elements, i.e.,*
$$B = \{X \subseteq A : |X| \text{ is even}\}.$$
Then, $|B| \leq 2^{n-1}$.

Question 1.4. 1. *(4 marks). Let* $a_1 := 1$ *and for every* $n \geq 2$*, let* $a_n := n - a_{n-1}$*. For every* $n \in \mathbb{N}, n \geq 1$*, prove that* $a_n \geq 1$*.*

2. *(6 marks). Write a summary of this course, preferably in the form of a story. Highlight your learnings. Your answer should be contained in at most two pages.*